A32NX air conditioning per the FCOM

The A32NX air conditioning controllers, alerts, STATUS lines and SD COND / BLEED pages follow the A320 FCOM (DSC-21-10, PRO-ABN-AIR, PRO-ABN-COND)

Aircraft	A32NX
Where to find it	Overhead AIR COND panel (PACK, HOT AIR, temperature selectors), ENG FIRE pushbuttons; E/WD; SD COND, BLEED and STATUS pages
Branch	`core/a32nx-air-conditioning`, based on `core/gps` (stacked: needs core/surveillance, core/a380x-mfd-fms and core/gps merged first; the STATUS lines use the A32NX STATUS page of core/gps).
Status	All four features tested in the sim by the user: OK. Two triggers cannot be produced in the sim (fourth duct excursion above 80 °C, pack valve position disagree): covered by Rust and vitest unit tests. A full MSFS restart is needed after installing (systems WASM).
Pictures	No pictures: the COND and BLEED pages are the stock FBW pages with the changes described.

1. Air conditioning controller per the FCOM
2. COND page trim air valve XX and air conditioning STATUS lines
3. Packs regulate cockpit and cabin separately with the hot air valve closed
4. BLEED page: each pack's own outlet temperature

1. Air conditioning controller per the FCOM

> **Aircraft:** A32NX | **Where:** Overhead AIR COND panel, ENG FIRE pb; SD BLEED / COND | **Status:** Tested in the sim by the user: OK: trim air valves, galley fan, FIRE pb per side, 30 s reopening. Fourth excursion and valve disagree: unit tests only.

What it is for

An audit of the air conditioning system controllers (ACSC) against A320 FCOM DSC-21-10 and PRO-ABN-AIR / PRO-ABN-COND found six deviations. All six are fixed:

- **ENG FIRE pushbutton:** each one closes the pack of its own side (DSC-21-10-50). Before, the pack flow controller read the ENG 1 FIRE pb for both packs: ENG 2 FIRE closed pack 1.
- **Pack valves after an engine start:** on the ground they reopen 30 s after the engine start (DSC-21-10-50 PACK pb: "reopening of the valves is delayed for 30 s to avoid a supplementary pack closure cycle during second engine start"); in flight at once.
- **Hot air valve and trim air valve failures:** the hot air pressure regulating valve closes when the cockpit trim air valve fails or both cabin ones fail, and stays open with one cabin trim air valve failed (DSC-21-10-20). Before, it closed on any trim air valve failure.
- **LAV + GALLEY FAN fault:** the cabin temperature sensors are no longer ventilated, so the FWD / AFT CABIN selectors set the cabin duct temperature directly (it was fixed at 15 °C). With ACSC 2 also lost, the cabin gets the cockpit duct air (PRO-ABN-COND).
- **DUCT OVHT** also triggers on the fourth excursion of a duct temperature above 80 °C in one flight, not only above 88 °C (PRO-ABN-COND). The count restarts at each takeoff run.
- **PACK 1(2) FAULT** also triggers when the pack flow control valve position disagrees with its command, not only on the loss of both lanes of the ACSC (PRO-ABN-AIR).

What you should see

Situation	What you should see
ENG 2 FIRE pb pushed	SD BLEED: pack 2 valve closes, pack 1 stays open (before: pack 1 closed).
Engine start on the ground, PACK pbs ON	The pack valves stay closed for 30 s after the start, then open.

Forward (or aft) cabin trim air valve failed	The hot air valve stays open; only that zone loses its trim.
Cockpit trim air valve failed, or both cabin ones	The hot air valve closes (SD COND: hot air valve symbol closed). The HOT AIR pb shows no FAULT for this automatic closure.
Lavatory and galley extraction fan failed	COND LAV+GALLEY FAN FAULT; turning the FWD / AFT CABIN selectors moves the cabin duct temperature directly.

Limitations and why

Limitation	Why	What to do
The fourth-excursion DUCT OVHT and the valve-disagree PACK FAULT cannot be produced in the sim.	There is no pack valve failure, and the valve closes by itself on low pressure; the duct temperature cannot be driven above 80 °C four times on purpose.	Covered by Rust tests and the AcscPackFaults vitest spec.
Not simulated: PACK REGUL FAULT, bypass valve and ram air inlet, pack overheat.	The packs are a placeholder in the FBW model.	None.
A full MSFS restart is needed after installing.	The controllers are in the systems WASM module.	Restart the sim.

How to use

1. Nothing new to set: use the AIR COND panel as usual (PACK 1 / 2, HOT AIR, COCKPIT / FWD CABIN / AFT CABIN).
2. In an engine fire the ENG FIRE pb closes the pack of that side only.

How to test

#	Do this	Expected result
1	A32NX on the ground, both engines running, PACK 1 and 2 ON. Open the SD **BLEED** page. Push the **ENG 2 FIRE** pb.	Pack 2 valve closes; pack 1 stays open. (Without `core/fire` the stock FBW also shows an ENG 2 FIRE warning when the pb is pushed: not part of this test.) Reset the pb.
2	Engines off, APU bleed ON, PACKs ON. Start engine 1 and watch the pack valves on the BLEED page.	The pack valves close during the start and reopen about 30 s after the engine is started.
3	Open the SD **COND** page. flyPad > Failures > ATA 21 Air Conditioning > **Forward Zone Trim Air Valve**.	The hot air valve stays open; the FWD valve shows amber XX (see the next feature).
4	Add **Aft Zone Trim Air Valve**.	The hot air valve closes (both cabin valves failed).
5	Clear them; fail **Cockpit Trim Air Valve** alone.	The hot air valve closes.
6	Clear the failures. flyPad > Failures > ATA 21 Air Conditioning > **Extraction Fan of lavatory and galley**. Turn the FWD CABIN selector from cold to hot.	E/WD: COND LAV+GALLEY FAN FAULT. The FWD cabin duct temperature on the COND page follows the selector directly.
7	Also fail **ACSC 2 Lane 1** and **ACSC 2 Lane 2**.	The cabin ducts take the cockpit duct temperature.

2. COND page trim air valve XX and air conditioning STATUS lines

> **Aircraft:** A32NX | **Where:** SD COND page; E/WD; SD STATUS page | **Status:** Tested in the sim by the user: OK.

What it is for

FCOM DSC-21-10-50 (COND page, zone trim air valve): "The arrow is green. It is replaced by amber crosses ("XX") if the valve fails". The FBW page had a TODO there. It now shows amber XX for a valve that either ACSC reports failed.

The air conditioning alerts now fill the ECAM STATUS page (PRO-ABN-AIR, PRO-ABN-COND). The STATUS page gains a limitations part (blue, before the information):

Alert	STATUS left column	INOP SYS
AIR PACK 1(2) FAULT	CKPT (CAB) AT FIXED TEMP when ACSC 1 (2) is lost	PACK 1(2), COND CTL 1(2) when the ACSC is lost
AIR PACK 1+2 FAULT	MAX FL.....100/MEA-MORA (blue, first), CKPT AT FIXED TEMP, CAB AT FIXED TEMP	PACK 1+2, COND CTL 1, COND CTL 2
AIR PACK 1(2) OFF	-	PACK 1(2)
AIR COND CTL 1(2)-A(B) FAULT	-	COND CTL 1(2)-A(B)
COND CKPT / FWD CAB / AFT CAB DUCT OVHT	CAB TEMP BY PACK ONLY	HOT AIR
COND HOT AIR FAULT	CAB TEMP BY PACK ONLY (hot air closed, packs on)	HOT AIR; PACK 1+2 with both packs off
COND L+R CAB FAN FAULT	-	L+R CAB FAN
COND LAV+GALLEY FAN FAULT	CAB TEMP CKPT CTL ONLY (ACSC 2 working) or CAB AT FIXED TEMP (ACSC 2 lost)	GALLEY FAN (+ PACK 2, COND CTL 2 with ACSC 2 lost)

- COND HOT AIR FAULT gets the full FCOM procedure: -HOT AIR...OFF, then ".IF HOT AIR STILL OPEN AND DUCT OVHT PERSISTS:" -PACK 1 OFF, -PACK 2 OFF, -DES TO FL 100/MEA-MORA, ".WHEN DIFF PR <1 PSI AND FL BELOW 100:" -RAM AIR ON, MAX FL 100/MEA-MORA (the descent lines in flight only).
- The TRIM AIR SYS FAULT line "CPKT TRIM VALVE" is now spelled **CKPT TRIM VALVE**.
- The FCOM FWD CRG HEAT line is left out: the FBW A320 has no cargo heating.

How to use

1. After an air conditioning alert, clear it with CLR and read the STATUS page (it comes up by itself, or press STS).
2. On the SD COND page, an amber XX in place of a zone arrow means that trim air valve has failed.

How to test

#	Do this	Expected result
1	A32NX powered, packs on. SD COND page. flyPad > Failures > ATA 21 Air Conditioning > **Cockpit Trim Air Valve**.	COND page: amber XX in place of the cockpit valve arrow. E/WD: COND TRIM AIR SYS FAULT, -CKPT TRIM VALVE (not CPKT).
2	Clear it. Fail **Cabin Fan 1** and **Cabin Fan 2**.	COND L+R CAB FAN FAULT, -PACK FLOW...HI. After CLR, STATUS: INOP SYS L+R CAB FAN.
3	Clear them. Fail **ACSC 1 Lane 1**.	AIR COND CTL 1-A FAULT; STATUS: INOP SYS COND CTL 1-A.
4	Also fail **ACSC 1 Lane 2**.	AIR PACK 1 FAULT; STATUS: CKPT AT FIXED TEMP (green), INOP SYS PACK 1 and COND CTL 1.

5	Also fail **ACSC 2 Lane 1** and **ACSC 2 Lane 2**.	AIR PACK 1+2 FAULT; STATUS: MAX FL.....100/MEA-MORA in blue on top, then CKPT AT FIXED TEMP and CAB AT FIXED TEMP; INOP SYS PACK 1+2, COND CTL 1, COND CTL 2.
6	Clear all. Fail **Cockpit Duct Overheat**.	COND CKPT DUCT OVHT; STATUS: CAB TEMP BY PACK ONLY, INOP SYS HOT AIR.
7	In flight, with **Trim Air Pressure Regulating Valve** failed (the valve stays where it is) and a duct overheat failure, watch COND HOT AIR FAULT if it is raised with the valve still open.	The lines .IF HOT AIR STILL OPEN AND DUCT OVHT PERSISTS:, -PACK 1 / -PACK 2 OFF, -DES TO FL 100/MEA-MORA, .WHEN DIFF PR <1 PSI AND FL BELOW 100:, -RAM AIR ON, MAX FL 100/MEA-MORA.

3. Packs regulate cockpit and cabin separately with the hot air valve closed

> **Aircraft:** A32NX | **Where:** Overhead HOT AIR pb and temperature selectors; SD BLEED | **Status:** Tested in the sim by the user: OK: HOT AIR OFF pack 1 16 °C / pack 2 29 °C; ON both back to the same value within 60 s.

What it is for

With the hot air pressure regulating valve closed (failure, HOT AIR pb OFF or the automatic closure), the trim air valves cannot mix the zones. FCOM DSC-21-10-40 (hot air valve failed closed): pack 1 regulates to the cockpit demand and pack 2 to the mean of the FWD and AFT cabin demands. With ACSC 2 lost, pack 2 follows the cockpit demand (PRO-ABN-COND). With the valve open, both packs still supply the lowest zone demand (DSC-21-10-20). Before, both packs always followed the lowest demand.

Limitations and why

Limitation	Why	What to do
The zone temperatures do not separate as much as the pack outlets.	The FBW air mixer gives every zone the same air, so pack 1 cannot cool the cockpit alone.	Read the effect on the BLEED page pack outlet temperatures.

How to use

1. With the HOT AIR pb OFF (or the valve failed closed), set the COCKPIT selector for the cockpit and the FWD / AFT CABIN selectors for the cabin: pack 1 serves the cockpit, pack 2 the cabin.

How to test

#	Do this	Expected result
1	A32NX, packs running, SD **BLEED** page. HOT AIR pb ON. Set COCKPIT cold, FWD and AFT CABIN hot.	Both pack outlet temperatures are the same (the lowest demand).
2	Set the HOT AIR pb OFF and wait about a minute.	Pack 1 outlet colder (cockpit demand), pack 2 warmer (cabin demand): for example 16 °C and 29 °C.
3	Set the HOT AIR pb ON again.	Both pack outlets go back to the same value within about 60 s.

4. BLEED page: each pack's own outlet temperature

> **Aircraft:** A32NX | **Where:** SD BLEED page | **Status:** Tested in the sim by the user: OK: HOT AIR ON 10 / 10 °C, OFF 15 / 30 °C.

What it is for

The pack outlet temperatures of the SD BLEED page were a placeholder showing the FWD cabin zone temperature on both packs. They now show the outlet temperature the air conditioning system computes for each pack, in 5 °C steps, amber above 90 °C (FCOM DSC-21-10-50, ECAM BLEED page, item 1).

Limitations and why

Limitation	Why	What to do
The pack bypass valve and the compressor outlet temperature are still placeholders.	The packs are not modelled internally yet.	None.

How to use

1. Open the SD **BLEED** page: the temperature above each pack is that pack's outlet temperature.

How to test

#	Do this	Expected result
1	A32NX, packs running, SD BLEED page, HOT AIR pb ON.	Both pack outlet values equal (for example 10), in steps of 5.
2	HOT AIR pb OFF, COCKPIT cold and cabin selectors hot.	The two values differ (for example 15 and 30): they are each pack's own outlet, no longer the FWD cabin temperature.